Research Project Presentation

On

Heart Disease Prediction Using Supervised Machine Learning Techniques

Submitted by

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Abstract

- Heart disease remains one of the leading causes of mortality worldwide, highlighting the importance of early diagnosis and risk assessment.
- Dataset: UCI Heart Disease dataset (Kaggle) with 303 patient records and 14 clinical features.
    - Features include: age, sex, chest pain type, resting BP, cholesterol, fasting blood sugar, ECG results, max heart rate, and more.
- Three supervised algorithms: Logistic Regression, Decision Tree, and Random Forest evaluated using accuracy, precision, recall, F1-score.
- Cross-validation, hyperparameter tuning (GridSearchCV, RandomizedSearchCV), and ROC-AUC analysis applied.
- Logistic Regression achieved the best performance: test accuracy ~88.5%, ROC-AUC ~0.91 after GridSearchCV tuning.
- Key predictors: chest pain type, resting ECG results, and ST segment slope identified via feature importance analysis.
- Keywords: Heart Disease Prediction, Machine Learning, Supervised Learning, Random Forest, Healthcare Analytics.
- Implemented in Python using Scikit-learn in a Jupyter Notebook/Google Colab

Introduction

- Heart disease is a leading global cause of death, responsible for millions of deaths annually.
- Early detection is critical — it significantly reduces mortality rates and improves patient outcomes.
- Traditional diagnostic approaches rely on clinical expertise and tests, which can be time-consuming and variable.
  - Misdiagnosis or delayed detection can lead to incorrect treatment and worsening of the patient's condition.
  - Not every patient has access to specialists, especially in remote or underserved regions.
- Machine learning has emerged as a powerful tool for predictive healthcare analytics and early diagnosis.
  - Supervised ML algorithms can analyze clinical parameters and predict disease risk automatically.
  - Logistic Regression and Random Forest offer efficient, interpretable models suitable for clinical decision support.
- Goal: Build an accurate ML-based prediction system for heart disease using clinical data from a public dataset.

Literature Review — Prior Work

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- **ML for Heart Disease Prediction:**
  - Logistic Regression and SVM have been widely used as baseline classifiers for heart disease prediction with accuracy up to 85%.
- **Ensemble Methods:**
  - Random Forest and XGBoost consistently outperform single classifiers, achieving accuracy of 87–90% on UCI dataset.
  - Gradient boosting approaches reduce overfitting and improve generalization on imbalanced clinical datasets.
- **Feature Selection Studies:**
  - Recursive feature elimination (RFE) and PCA have been used to identify the most informative clinical features.
- **Hyperparameter Tuning:**
  - GridSearchCV and RandomizedSearchCV have been shown to significantly boost model performance over default settings.
  - Cross-validation with ROC-AUC analysis is a standard practice for reliable model validation in medical prediction tasks.

Literature Review — Recent Advances

- Recent studies using deep learning (ANN, LSTM) on ECG data have achieved accuracy exceeding 90% for cardiac event prediction.
- Federated learning approaches have been explored for privacy-preserving heart disease prediction across multiple hospitals.
- Hybrid CNN + classical ML pipelines have demonstrated strong performance by combining image/signal data with structured clinical features.
- **Benchmark Observations:**
  - The UCI Cleveland dataset remains the most widely used benchmark; Random Forest and Logistic Regression are top performers.
  - SMOTE-based oversampling techniques have been applied to address class imbalance in heart disease datasets.
- **Key Observations from Literature:**
  - Hyperparameter tuning, cross-validation, and feature importance analysis consistently improve model reliability.
  - Interpretability (feature importance, SHAP values) is increasingly important for clinical adoption of ML models.
  - These findings motivate our comparative evaluation of Logistic Regression, Decision Tree, and Random Forest.

Our Contributions

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- Research Question: Which supervised ML algorithm best predicts heart disease from clinical parameters?
- **Contribution 1: Dataset Preparation**
  - Sourced and preprocessed the UCI Heart Disease dataset (Cleveland) with 303 records and 14 clinical features.
  - Applied 80/20 train-test split with standard feature scaling for fair model evaluation.
- **Contribution 2: Model Implementation and Tuning**
  - Implemented Logistic Regression, Decision Tree, and Random Forest using Scikit-learn with default and tuned parameters.
  - Applied GridSearchCV and RandomizedSearchCV for optimal hyperparameter selection (e.g., $C \approx 0.204$ for Logistic Regression).
- **Contribution 3: Comprehensive Evaluation**
  - Evaluated all models using accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix, and 5-fold cross-validation.
  - Identified clinically meaningful feature importance (chest pain type, ECG results, ST slope) to support interpretability.

Methodology — Dataset Description

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- Dataset: UCI Heart Disease dataset (Cleveland) obtained from Kaggle.
- **Dataset Attributes:**
  - 303 patient records with 14 clinical features including age, sex, chest pain type (cp), resting BP (trestbps), cholesterol (chol), fasting blood sugar (fbs), resting ECG (restecg), max heart rate (thalach), exercise-induced angina (exang), ST depression (oldpeak), slope, major vessels (ca), thalium stress result (thal), and binary target variable.
  - Target variable: binary — presence (1) or absence (0) of heart disease.
- **Data Split — 80/20 strategy:**
  - 80% (242 records) used for training the models.
  - 20% (61 records) reserved for final testing and evaluation.
- This split mirrors real-world conditions where unseen patient data is used for final model assessment.
- Fixed random seed (42) applied to ensure reproducibility. No missing values were found in the dataset.

Data Preprocessing

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- Feature Scaling: Standard scaling applied to normalize all 14 clinical features for consistent model input.
- Train-Test Split: 80/20 split using train_test_split with random_state=42 for reproducibility.
- **Missing Value Check:**
  - Dataset verified to contain no missing values — no imputation required.
  - All 303 records retained for model training and evaluation.
  - Feature Encoding: All categorical and binary features already encoded numerically — no additional transformation needed.
- **Purpose of Preprocessing:**
  - Scaling prevents features with large ranges (e.g., cholesterol, BP) from dominating distance-based computations.
  - Consistent preprocessing ensures fair comparison across Logistic Regression, Decision Tree, and Random Forest.
  - Reproducible splits with fixed random seed enable reliable benchmarking and result comparison.
- All preprocessing steps implemented using Scikit-learn's StandardScaler and train_test_split utilities.

Logistic Regression Model

- A linear probabilistic classifier suitable for binary classification tasks.
- **Feature Importance:**
  - Solver: lbfgs | Max iterations: 1000 | Random state: 42 for reproducibility.
  - Hyperparameter tuning: GridSearchCV used to identify optimal C ≈ 0.204 over a log-scale grid.
- **Post-Tuning Performance:**
  - Test Accuracy: ~88.5% | Cross-Validated Accuracy: ~85.2% (5-fold).
  - Precision: ~84% | Recall: ~88% | F1-Score: ~86% | ROC-AUC: ~0.91.
  - Logistic Regression was selected as the final model due to highest overall performance.
- **Feature Importance:**
  - Feature importance derived from coefficients: chest pain type (cp), resting ECG (restecg), and ST segment slope are strongest predictors.
  - Sex showed the lowest predictive contribution among all 14 features.

Random Forest Classifier

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- An ensemble of decision trees that aggregates predictions to improve accuracy and reduce overfitting.
- **Training Configuration:**
    - n_estimators, max_depth, min_samples_split, and min_samples_leaf tuned via RandomizedSearchCV with 5-fold CV.
    - Default Random Forest instantiated first; then tuned using randomized search over parameter distributions.
    - Random state: 42 for consistent results across runs.
- Post-Tuning Performance:
- **Test Accuracy: ~83.6% | Competitive performance after RandomizedSearchCV tuning.**
    - Precision: ~82% | Recall: ~83% | F1-Score: ~83% | ROC-AUC: ~0.89.
    - Native Feature Importance:
    - Random Forest provides built-in feature importance scores — chest pain type and thalium stress result ranked highest.
    - Slightly lower accuracy than Logistic Regression, but provides complementary insights through ensemble voting.
- Random Forest is less interpretable than Logistic Regression but more robust to

Training Configuration — All Models

- All three classifiers trained under comparable conditions to ensure a fair and consistent comparison.
- **Common Settings:**
  - Dataset: UCI Heart Disease (303 records, 14 features) | Train-Test Split: 80/20 | Random State: 42.
  - Scaling: StandardScaler applied to all features before training.
  - Evaluation Metrics: Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix.
  - Cross-Validation: 5-fold CV used for all models to compute generalized performance metrics.
  - Environment: Jupyter Notebook / Google Colab | Framework: Scikit-learn | Language: Python.
- **Key Differences Between Models:**
  - Logistic Regression: linear model, fast training, tuned via GridSearchCV (optimal C ≈ 0.204).
  - KNN: distance-based, manually tuned n_neighbors (1–20); best score ~75% — lowest performer.
- Random Forest: ensemble method, tuned via RandomizedSearchCV over n_estimators, max_depth, min_samples settings.

Results — Logistic Regression Performance

- Test Accuracy: ~88.5% — best performance among all three classifiers.
- Cross-Validated Accuracy: ~85.2% (5-fold) — confirms strong generalizability.
  - Precision: ~84% | Recall: ~88% | F1-Score: ~86% — well-balanced across positive and negative classes.
  - ROC-AUC: ~0.91 — strong discriminative ability between heart disease positive and negative cases.
- **Confusion Matrix Analysis:**
  - Visualized using Seaborn heatmap — shows accurate separation of true/false positives and negatives.
- **Feature Importance (from Logistic Regression Coefficients):**
  - Strongest positive predictors: chest pain type (cp), resting ECG (restecg), ST segment slope.
- **Weakest predictor: sex — lowest contribution to model output.**
  - ROC Curve plotted to confirm model's ability to discriminate across all classification thresholds.
  - Logistic Regression selected as the final recommended model due to highest overall metrics.

Results — Random Forest Performance

- Test Accuracy: ~83.6% — competitive performance after RandomizedSearchCV hyperparameter tuning.
- **Hyperparameter Tuning Results:**
  - Best parameters identified via RandomizedSearchCV: optimal n_estimators, max_depth, min_samples_split, min_samples_leaf.
  - 5-fold cross-validation applied to confirm parameter selection and generalizability.
- Precision: ~82% | Recall: ~83% | F1-Score: ~83% | ROC-AUC: ~0.89.
- Feature Importance Analysis:
- Random Forest provides native feature importance scores — chest pain type and thalium stress test ranked as top predictors.
- **Challenges:**
  - Slight misclassification in borderline cases where atypical chest pain features overlap with normal presentations.
  - Performance slightly lower than Logistic Regression — likely due to the small dataset size (303 records).
- Random Forest remains valuable for its native interpretability and robustness to non-linear feature interactions.

Comparative Analysis

- Test Accuracy: Logistic Regression = ~88.5% vs. Random Forest = ~83.6% — Logistic Regression wins.
- ROC-AUC: Logistic Regression = ~0.91 vs. Random Forest = ~0.89 — Logistic Regression shows better discrimination.
- Training Speed: Logistic Regression faster — simpler model; Random Forest slower due to ensemble building.
- Interpretability: Logistic Regression — linear coefficients; Random Forest — feature importance scores; both interpretable.
- Tuning Method: Logistic Regression via GridSearchCV; Random Forest via RandomizedSearchCV.
- Overfitting: Logistic Regression — lower tendency; Random Forest — moderate tendency, controlled via max_depth.
- Classification Quality: Both strong; Logistic Regression marginally better on all key metrics.
- KNN Comparison: Manually tuned to best k (~75% accuracy) — significantly underperforms both other models.

- Conclusion: Logistic Regression outperforms Random Forest on this dataset —

Conclusion

- Compared three supervised ML classifiers for heart disease prediction: Logistic Regression, Decision Tree, and Random Forest.
- Logistic Regression achieved the best performance (test accuracy ~88.5%, ROC-AUC ~0.91) after GridSearchCV tuning.
- Hyperparameter tuning proved critical — GridSearchCV identified optimal C ≈ 0.204, significantly improving Logistic Regression.
- Random Forest also performed competitively (~83.6%) and benefited from RandomizedSearchCV tuning with native feature importance.
- **Practical Recommendation:**
  - Logistic Regression is the preferred model for deployment — interpretable, fast, and most accurate on this dataset.
  - Even a 3–5% accuracy improvement over baseline can be clinically significant in heart disease risk screening.
- Key Insight: Chest pain type, resting ECG results, and ST segment slope are the most influential clinical predictors.
- Future Work: Expand dataset, explore deep learning (ANN, neural networks), and develop a real-time web/mobile risk screening interface.

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Thank You

- Thank you for your attention.
- Questions and feedback are welcome.